1DV701 Assignment 2 - Web Server

The second assignment is designed to familiarise you with the Java networking API. Your task is to develop a web server using Java that can serve static content. This assignment should improve your understanding of TCP, server sockets, and the HTTP protocol.

Note: This assignment is performed in groups of two. Groups of three or larger are not allowed

Reference material

- HTTP overview
- Java networking tutorial
- Official Java documentation

Useful downloads

- PuTTY client (for Windows users).
    - If you use Homebrew (on macOS), you can use: brew install telnet
    - You can also use nc instead of telnet on either platform.
- Insomnia REST client or Postman REST client - REST clients for testing
- POST tldr.sh – Community-driven cheat-sheets for Unix commands

Problem 1

Implement a simple web server that accepts multiple client connections on some port (e.g., 8888) using the ServerSocket class. The server should serve (at least) HTML and PNG files from a local directory, using the GET method in the HTTP protocol.

The response from the server should include *correct and relevant headers* as well as the requested file.

The server should run until you terminate it (ctrl-c or ctrl-d keyboard shortcut).

***Program requirements*:**

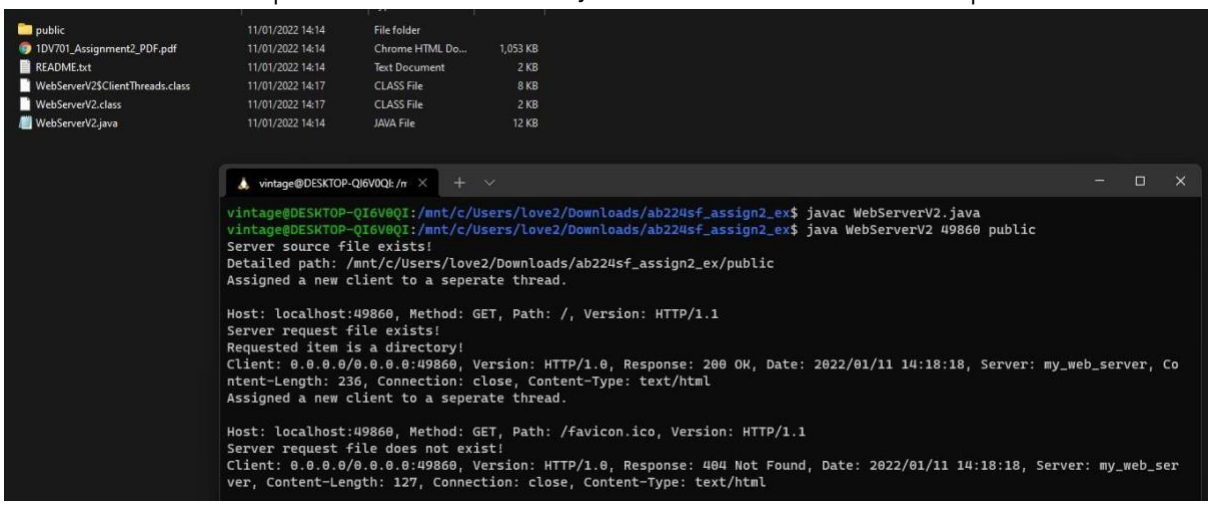
- Two program arguments in the specified order
    - The port (listening port)
    - Relative path to /public directory. (Do **NOT** use absolute paths)
- You are *not* allowed to use third-party libraries or existing Java classes (such as com.sun.net.httpserver) since these make the problem trivial.
- Console output from the program should provide useful debugging information and, in general, make sense. Eg, connections being established and content served are a good idea to display.
- Use the provided public folder.
- Use the provided Python test script to check if your implementation works as expected.
- Use thread pooling instead of creating new threads for each connection.

Note: What is the /public directory?
Think of this folder as the serving directory from which our web server will serve html files and images.

Accessing http://yourip:port/stuff.html should serve the file from your_project_folder/public/stuff.html.

Note: Using the command line
Your assignment is going to be tested via the command line; it will not be run via an editor/IDE. Note that this means you should verify that everything works correctly via the command line. If you are on Windows, WSL is recommended to use here to get into a Linux environment for more convenient testing. Java also exists for powershell/cmd. Compiling is done via the command javac, while the actual class is run using the java command.

Note: An example of someone doing it right
The following image represents someone who has fully completed this problem (and a small part of P2) and has provided all the parts necessary to get full marks. Take note of the document structure of the submission (visible in the file explorer) and the way compiling and running the solution works. Take special note of where the .java main file is in relation to the public folder.



How to start:

Start by implementing a server that accepts connections and returns a predefined (valid) HTTP/HTML response. Once you can connect to the server with a web browser and view the predefined HTML page, you should add support for reading HTML files from disk and returning them.

Finally, add support for images (binary files). If a user requests a specific directory, e.g. / or user1/pages, and this directory contains an index.html file, then this file should be returned. Note that the ending / is optional, so http://myserver.com/user1 and http://myserver.com/user1/ should both return the index.html file in the user1 directory.

Prepare a few HTML and PNG files in a directory hierarchy (e.g. /memes/rarepepe.png) and use this to test your web server. Make sure to include both index.html files as well as other named HTML files, e.g., myfile.html. Your report should include screenshots of the browser window when you request a named HTML page, an image, and a directory (with an index.html file).

Your web server should be robust -- you have to handle possible exceptions. **Document which exceptions you handle and why.** Do NOT allow a user to escape the public directory and access other files on the computer.

Problem 2

The web server you implemented in Problem 1 only supports the HTTP response 200 OK. You should now add support for the following response types:

- 302
- 400
- 404

When implementing 302, hardcode a specific url that redirects to any resource of your choosing. Include a description of how you tested each of these response codes with screenshots of the result in your report. Include in the instruction text file how to trigger each of these.

Problem 3

After completing Problems 1 and 2, you should configure the web server to handle a simple login page. You need to create a new HTML file with a form that handles the POST action. You also need to have an external file which contains the login information, it can be, for example, a JSON or txt. If the received information is correct, then the server should send 200 OK; if not, then 401 Unauthorised.

Add support for the HTTP POST method. Add a webpage with a form that allows a user to upload PNG images somewhere in your directory hierarchy. We recommend implementing a multipart-form parser.

Point distribution for the programming part

Prerequisite for getting points:

- **That the program compiles and runs.**
- **Both students in the group have made a reasonable number of commits spread out over several weeks.**

Some items are marked with a "*" and bold text; these are mandatory to pass the assignment, in addition to getting enough points

P1:

- **Working HTTP server with HTML+HTM support - 3p ***
- Thread pooling for connections - 4p
- Image support - 2p
- Safety against escaping the root - 2p
- **Exception handling - 2p ***
- **Program arguments in correct order - 1p ***
- Console output - 1p

P2:

- 302 Implementation - 2p
- **404 Implementation - 1p ***
- 400 Implementation - 2p

P3:

- **HTTP POST login - 3p ***
- HTTP POST Implementation with a supporting webpage for image upload feature - 4p

Getting the points for the mandatory task will give you an E. Getting full points will give an A. Other grades are distributed between these.

Points are deducted from the total for things such as poor code quality or other strange things with your program or version control, so even if you do all the mandatory tasks, you could still get an F or Fx.

Submission

All submissions are done in GitLab. To submit, please create a new **issue** in your repository, add the label **Submitted**, and select the TA for your group as an **Assignee** when prompted:

Your repository should contain the following:

- A **PDF** report containing a brief overview of each problem, along with the things specifically outlined to write about.
- All **.java** files required to run the program.
- A public folder containing resources.
- A **README.MD** file containing instructions on how to run the program

Make sure to have a good Git **commit history** to demonstrate your collaboration. Make the commit names meaningful! Example guide:
https://www.freecodecamp.org/news/how-to-write-better-git-commit-messages/

Please agree on the final submission before you submit.

In the **README**, include a summary of the instructions on how to run the program.

In the **report**, include how much each of the participants of the group contributed and how you split the workload (in percent). For example, student_name_1: 45%, student_name_2: 55%. If these differ too much, the participants might receive different grades for the assignment. If one of the members in the group does not contribute at an acceptable level, please notify the teacher as soon as possible.